RELATIVE PSEUDO-COHERENCE IS PRESERVED UNDER TOR-INDEPENDENT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17111-17120 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17122-17152, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author writes O_{X_, x} with a stray underscore in the displayed formula; retained verbatim from source lines 17133-17145.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The affine-space characterization of relative pseudo-coherence (same chapter, lemma-relative-pseudo-coherence-characterize implicitly via the companion result) and pullback of pseudo-coherent complexes (Cohomology, lemma-pseudo-coherent-pullback).

Source reference. Original lemma 09U7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17152–17152. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08HY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17152–17152. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $m \in \mathbf{Z}$. Let E be an object of $D(\mathcal{O}_X)$ which is m-pseudo-coherent relative to S. Let $S' \to S$ be a morphism of schemes. Set $X' = X \times_S S'$ and denote E' the derived pullback of E to X'. If S' and X are Tor independent over S, then E' is m-pseudo-coherent relative to S'.*

Proof. The problem is local on X and X' hence we may assume X, S, S', and X' are affine. Choose a closed immersion $i : X \to \mathbf{A}^n_S$ and denote $i' : X' \to \mathbf{A}^n_{S'}$ the base change to S'. Denote $g : X' \to X$ and $g' : \mathbf{A}^n_{S'} \to \mathbf{A}^n_S$ the projections, so $E' = Lg^*E$. Since X and S' are tor-independent over S, the base change map (Cohomology, Remark 08HY) induces an isomorphism

$$Ri'_*(Lg^*E) = L(g')^*Ri_*E$$

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Namely, for a point $x' \in X'$ lying over $x \in X$ the base change map on stalks at x' is the map

$$E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{\mathbf{A}^n_S,x}} \mathcal{O}_{\mathbf{A}^n_{S'},x'} \longrightarrow E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}_{X',x'}$$

coming from the closed immersions i and i'. Note that the source is quasi-isomorphic to a localization of $E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{S,s}} \mathcal{O}_{S',s'}$ which is isomorphic to the target as $\mathcal{O}_{X',x'}$ is isomorphic to (the same) localization of $\mathcal{O}_{X,x} \otimes^{\mathbf{L}}_{\mathcal{O}_{S,s}} \mathcal{O}_{S',s'}$ by assumption. We conclude the lemma holds by an application of Cohomology, Lemma 09U7. $\square$